

Karnataka Storage Decision Brief

Banking, General Network Access and BESS economics under KERC's draft Regulations of June 2026

Report No. 1

D. S. Nikam *Independent research — energy storage and power markets*

Version 1.3 · Published 31 August 2026

What changed since v1.2 — a substantive correction.

The full 167-page judgment in *Fortune Five Hydel Projects Pvt. Ltd. v. KERC* (APTEL, Appeal No. 42 of 2018 and batch, 29 March 2019) has now been read. **Version 1.2 wrongly declined to attribute natural justice and promissory estoppel to the Tribunal**, on the footing that those doctrines appeared only in the appellants' submissions. They do appear there — but the Tribunal also **framed them as Issue No. 2 of its own** (p. 129) and **held on all three grounds** at paras 15.14 and 17.3 before setting the order aside.

Section 6.6.3.4 now sets out the holding from the primary judgment, with page and paragraph references. Section 6.6.3.5 states it narrowly, and **6.6.3.6 records the error and the rule that follows from it**. The Annexure has been updated accordingly.

The error was caught by a reader within hours of publication. No figure, tariff or regulation reference in this brief has changed — the correction is confined to the account of one judgment.

© 2026 D. S. Nikam. Free to share with attribution. Please do not alter, excerpt selectively, or republish as your own work.

This is independent research. It is not legal advice, not investment advice, and not a recommendation to procure any battery, equipment or service. Regulatory treatment and project economics must be checked against the final notification and against project-specific documents.

Contents

Section 0 — The decision clock **Section 1** — What happens to your non-solar hours **Section 2** — The three structural routes **Section 3** — The slot trap in banking **Section 4** — The cost of access **Section 5** — Inconsistencies in the draft **Section 6** — What other States are doing **Annexure** — Method, sources and limitations

The new Connectivity / GNA Regulations and the ending of banking

Sections 0 and 1

Position as at September 2026

Before you read

This brief is based on the **draft** Karnataka Electricity Regulatory Commission (Connectivity and General Network Access to the Intra-State Transmission and State Distribution System) Regulations, 2026 — Notification No. KERC/F-32/V-27/2083 dated 4 June 2026, published in the Karnataka Gazette on 11 June 2026. **Regulation numbers and dates may change in the final notification.**

Every figure carries its source, clause and date. Where verification is outstanding, it is stated. This is not legal advice.

On two terms

In this brief, **"access right"** means the regulatory permission, under a sanctioned Connectivity or General Network Access ("GNA"), **to inject energy into the grid outside solar hours**. It does not mean the right to draw down banked energy — that is a separate question, and it is dealt with in Section 3.

Every operative conclusion carries a status: **Verified** (express in the Gazette or in a KERC order) · **Conditional**

(dependent on project approvals, the GNA agreement or SLDC procedure) · **Interpretive** (drawn from the wording of the regulation; clarification required from the Commission or the SLDC) · **Unknown** (no public data available).

0 | The decision clock

Date	What happens	Regulation
1 October 2026	The Regulations commence. The Open Access Regulations, 2025 stand repealed.	1.2, 54.1
1 October 2026 - 31 December 2026	The only window in which existing solar entities may apply for additional capacity or for non-solar hour access	Provisos to 5.10(b), 5.20(b)
~ first week of January 2027	Every solar entity that has not applied is converted to solar hour access	5.10(b), 5.20(b)
The renewal date of your WBA	Banking ends for holders of the older agreements	Proviso to 39.1
31 March 2030	Monthly banking closes for everyone	39.1

The first three rows are common to all and turn on three months. The fourth turns on your own Wheeling and Banking Agreement — and for many it falls before the fifth.

The remainder of this brief explains those five rows, and what can be done about them.

1 | What happens to your non-solar hours

1.1 What changes

Under the present regime, the network access granted to a solar project runs for the whole day. The project does not inject at night because it has no energy at night — but it holds the **right**.

The new Regulations divide that right in two: **solar hour access** and **non-solar hour access**.

Regulations 5.10(b) and 5.20(b) provide that **every** solar-based Renewable Energy Generating Station ("REGS"), Renewable Hybrid Generating Station ("RHGS") with a solar component, or Renewable Power Park Developer ("RPPD") that has been intimated in-principle or final grant of Connectivity, or whose GNA is effective, **shall be converted** into an entity with solar hour access **within one week after the expiry of three months** from the date the Regulations take effect.

What remains after conversion:

- **During solar hours** — injection scheduling rights for the full quantum of Connectivity, as before
- **During non-solar hours** — injection **limited to the capacity of the non-solar source**

For a pure solar project, that second figure is nil.

1.2 What that means in practice

After conversion, your solar project **cannot** inject into the grid in the evening — even if it has the energy to inject.

This is not a question of tariff. **It is a question of access.** However attractive the price, without the right there is no transaction.

And it is distinct from the slot restriction on banking. That restriction is commercial; this one is physical. (*Banking is dealt with in Section 3.*)

1.3 To whom it applies

Applies	Does not apply
Solar REGS	Wind REGS
Solar-based RHGS	Standalone ESS $\geq$ 5 MW
Renewable Power Park Developers	

Under Regulation 4.4, a wind-based REGS **or an Energy Storage System ("ESS")** of 5 MW and above may elect between **non-solar hour access** (under Regulation 5.10) and **full day access**.

Regulation 4.4 does not extend that election to a pure solar project. **Adding an ESS to the same project opens it.**

Three things should nevertheless be kept apart:

	What it is
Eligibility	Regulation 4.4 confers eligibility to apply for non-solar hour access as an ESS
Technical capability	A battery can charge in solar hours and discharge in non-solar hours
Actual entitlement	Depends on sanctioned Connectivity, effective GNA, metering, scheduling and available network capacity

That the Regulations recognise ESS as a class does not mean that every solar project automatically obtains full day access. (Conditional)

1.4 The window

The provisos preserve a route of defence. Within the same three months — **1 October 2026 to 31 December 2026** — an application may be made under Regulation 5.2 (adding capacity within existing Connectivity) or Regulation 5.10(a) (independent non-solar hour access), and in carrying out the conversion the Nodal Agency **shall consider** that application.

If no application is made, the conversion takes effect automatically.

The draft contains no provision for extension of that window.

1.5 An internal inconsistency — worth recording

Both provisos carry an exception: if the non-solar capacity that would be released exceeds a stated threshold, the conversion does not take place.

But in the two places the condition is written **in opposite directions**:

Clause	Condition
Proviso to 5.10(b)	No conversion if the capacity that can be made available for non-solar hours is less than 5 MW
Proviso to 5.20(b)	No conversion if that capacity is more than 5 MW

The same inconsistency appears between Regulations **4.5** (transmission) and **4.8** (distribution).

The intended logic appears to be this: **if no meaningful capacity is released for another entity, there is no purpose in withdrawing the full-day right from the original entity**. On that logic, both 4.5 and 5.20(b) have been written the wrong way round.

This is not a matter of opinion. Two clauses give contradictory answers on identical facts. It is to be expected that this will be corrected before the final notification — and which version survives determines the position of projects at the margin.

(The full six-clause map, and the third departure at 5.20(c), are set out in Section 5.)

1.6 This is not confined to Karnataka

The structure derives from the Central Electricity Regulatory Commission's **Connectivity and General Network Access (Third Amendment) Regulations, 2025**, which divided GNA for all renewable projects into solar and non-solar hours. The reason is on record: transmission capacity allotted to solar projects lies unused when solar is not available.

At State level, Karnataka is **the first**. Maharashtra's new renewable energy and energy storage policy (Government Resolution dated 18 March 2026) records the same possibility at Clause 6.5(C), but there the STU and MSLDC are directed to **study it over twelve months**.

The Karnataka experience in January 2027 is therefore an early warning for projects in other States.

1.7 One limitation

This brief does not state **how much capacity** the conversion will affect.

The Regulations say "every" solar REGS / RHGS / RPPD — that is, not merely those holding open access or banking, but everyone holding Connectivity or GNA. That set includes projects contracted to the distribution licensees, and no consolidated list of them is publicly available.

In Section 3 we give a figure for the capacity that loses banking — **3,725 MW**. That is not an estimate of all solar capacity in Karnataka, nor of all capacity affected by Regulations 5.10(b) / 5.20(b).

It is the measure of one identifiable subset within the available banking dataset — not the size of the market, but a verifiable floor. (Verified, but a subset)

1.8 Four distinct layers — to avoid confusion

Four things are discussed in this brief. They should not be run together:

Layer	The question it answers	The instrument
1. Connectivity / GNA	When may I inject into the grid?	Draft Regulations 2026 — the subject of Section 1
2. Banking	When may the injected energy be taken back, and at what cost?	Regulation 39 — the subject of Section 3
3. ToD tariff	What is that energy worth in rupees at that hour?	The Tariff Order — the subject of Section 3
4. ESS	The physical means of moving energy across time	Not a regulatory entitlement — an asset

The first three are regulatory layers. The fourth is technical. Until the first is settled, discussion of the other three is idle — without the right, neither the tariff nor the asset is of any use.

The order of this brief follows that sequence: entitlement first, then settlement, then price, then the asset.

Section 2 — The three structural routes

Position as at September 2026

● **Based on the draft.** Every regulation number in this section is taken from the Commission's **draft** (Connectivity and GNA) Regulations, 2026 — Notification KERC/F-32/V-27/2083 dated 4 June 2026, published in the Karnataka Gazette on 11 June 2026 (Volume 161, Issue 112). The final notification, the commencement date and the STU / SLDC detailed procedures are all awaited. **None of these routes is an operative entitlement today.**

Status marks used: **Verified** · **Conditional** (the provision exists, but is subject to contractual, technical or approval conditions at project level) · **Interpretive** (a conclusion drawn from the text) · **Unknown**

2.0 These are not recommendations

Section 1 showed that solar entities which have not applied by January 2027 will be converted automatically to solar hour access.

The draft contains **three routes** against that outcome. They are options, not remedies — each carries its own condition, its own cost and its own uncertainty. For each, the same three questions are asked below: **when it works, when it does not, and what to check.**

2.1 Route A — adding storage within existing Connectivity

Regulation 5.2 (transmission) and **5.12** (distribution)

What it provides

With the prior approval of the Nodal Agency, a generating station or an ESS may add generation capacity or an Energy Storage System **within the quantum of Connectivity already granted to it — including capacity owned by any other entity.**

Conditions: the existing Connectivity grantee remains responsible for compliance with the Grid Code and the other regulations, as "Lead Generator" or "Lead ESS"; and **net injection at any point of time must not exceed the total quantum of Connectivity granted.**

*This text is carried across verbatim from the Forum of Regulators' model regulations. **Verified** — which also means it is not peculiar to Karnataka; it will appear in other States.*

Why it matters

Because the regulation has settled the question of whose capital is at risk.

A solar developer without the funds for a battery can still preserve its non-solar hour rights — a third party's battery sits inside the developer's own Connectivity. No fresh application, no additional bay, no new line.

This opens the entire commercial structure of storage-as-a-service.

When it does not work

- If the sanctioned quantum of Connectivity is already fully utilised, there is no headroom
- Prior approval of the Nodal Agency is required, and **the draft prescribes no time limit for it (Unknown)**
- Lead Generator responsibility stays with the solar developer whoever owns the battery — the regulatory consequences of a third party's error fall on the developer

What to check

Conn-BG3 is not one figure. It is two. The draft sets it differently for two distinct situations:

Situation	Conn-BG3	Regulation
Enhancement of capacity, augmentation without ATS	₹1 lakh per MW	8.2(b) transmission · 8.6(d) distribution
Enhancement of capacity, augmentation with ATS	The full estimated cost of the ATS	8.3(f) transmission · 8.7(f) distribution

The general proposition "Conn-BG3 = ₹1 lakh/MW" is therefore wrong. Where an Associated Transmission System is required, the figure is not proportionate to the size of the project — it is proportionate to the network works, and it can be very much larger. (Verified)

And how long that money is tied up: Regulation 16.2 — Conn-BG2 and Conn-BG3 are returned **in five equal parts over five years**, corresponding to capacity declared under commercial operation. Conn-BG1 is returned within thirty days of declaration of commercial operation of the full capacity (16.1).

Neither figure appears in most IRR models — the full cost of the ATS bank guarantee, and its being tied up for five years. On a 100 MW project this is working capital in crores.

Keep MW and MWh distinct. Regulation 5.5 (standalone ESS) and Regulation 5.3 (REGS with storage) both require the quantum of Connectivity to be applied for **in MW** — maximum injection or maximum drawal, whichever is higher. **Not in MWh.** Moving from a two-hour to a four- or six-hour duration therefore does not increase the quantum of Connectivity or its cost. (Verified)

2.2 The core of Route A — eight questions

The regulation says only that the battery may be owned by another entity. **It does not say how that contract is to be built.** And that is where the transaction stalls.

If the eight questions below are not answered, Regulation 5.2 remains merely a proposition that "*capital is saved*". If they are answered, it becomes a **bankable structure**.

#	Question	The document that must capture it
1	Who owns the ESS?	Asset ownership and security documents
2	Who controls dispatch?	EMS / dispatch protocol and operating agreement
3	Who holds title to, and the attributes of, the charging energy?	Energy accounting, REC / attribute and settlement clauses
4	Who bears deviation / DSM risk?	Scheduling and indemnity clauses
5	Is Connectivity separate or shared?	Connectivity / GNA grant and interconnection agreement
6	Who pays for degradation and augmentation?	Long-term O&M, availability guarantee, lifecycle reserve
7	How does the solar developer obtain firm delivery?	Delivery clause of the tolling agreement / ESA / PPA
8	What step-in and security does the lender have?	Assignment, direct agreement, escrow, step-in

Question 4 is the one most often overlooked. The proviso to Regulation 5.2 is explicit: **responsibility for Grid Code compliance rests with the existing Connectivity grantee** — that is, with the solar developer. If the battery is scheduled badly, the deviation charges arise in the developer's name. Unless the contract passes that exposure back to the battery owner, the developer is carrying the risk of another party's asset for nothing.

And without Question 8 no lender will proceed. The battery belongs to someone else, the Connectivity belongs to you, and what the lender takes possession of on default must be written down — otherwise the structure stays on paper.

These are not points of legal advice; they are points of diligence. Each requires its own answer.

2.3 Route B — independent non-solar hour access

Regulation 5.10(a) (transmission) and **5.20(a)** (distribution)

What it provides

A wind-based REGS (with or without storage) **or an ESS** may seek Connectivity with non-solar hour access, in one of two places:

(i) through a separate dedicated line, **or (ii)** at a terminal bay **already allocated to another REGS or Renewable Power Park with solar hour access**.

(For transmission these are 5.10(a)(i) and (ii); for distribution the same provision appears at 5.20(a), where the sub-clauses are numbered (iii) and (iv) — a numbering error in the draft, not a difference of substance.)

Why it matters

Limb (ii) is being overlooked — and it is not an inference.

The text of Regulation 5.10(a) describes the terminal bay in terms: "*which is already allocated to another REGS or Renewable Power Park, with solar hour access.*"

A solar project's evacuation infrastructure is used in the daytime and **stands idle at night**. The draft permits, in express words, another entity to seek non-solar hour access at that same bay.

The uncertainty is not in the text but in implementation — the commercial agreement, the Nodal Agency's approval, and available capacity. (Verified; implementation Conditional)

And a substantial saving — subject to a condition

Proviso to Regulation **8.2(a)**: where Connectivity is sought at a terminal bay constructed or being constructed by another Connectivity grantee, **Conn-BG2 is not required**.

But there is a condition: the bay must have been constructed by that other grantee **at his own cost**. The exemption does not apply to a bay built by the licensee. The corresponding provision for distribution is at 8.6(a).

In practice this condition is selective — it determines which bays are worth approaching. (Verified)

When it does not work

- **An agreement with the existing bay holder is required, and he may refuse**
- Residual capacity must be available in the bay and in the line (Regulation 9.2)
- The Conn-BG2 exemption applies only to a bay built at the grantee's own cost

The 5 MW threshold is not a barrier — it determines the level. Regulation 5.10(a) applies at transmission for **≥ 5 MW**, and Regulation 5.20(a) at distribution for **< 5 MW**. A smaller project is therefore not denied Route B; **it moves to the distribution level** — a different Nodal Agency, different fees, and the framework of Regulation 4.1(B).

The 5 MW inconsistency — the picture is larger than it appears

The draft uses the 5 MW threshold in **six places**. The direction is not consistent:

	Transmission	Distribution
Eligibility (4.5 / 4.8)	≥ 5 MW → full day	< 5 MW → full day
Conversion (5.10(b) / 5.20(b))	< 5 MW → full day	> 5 MW → full day
Fresh application (5.10(c) / 5.20(c))	≥ 5 MW → solar hour	< 5 MW → solar hour

On the transmission side, 5.10(b) and 5.10(c) are consistent with one another and consistent with the Forum of Regulators text — **4.5 alone runs the other way**. On the distribution side the position is exactly reversed: **4.8 matches the FoR text, and both 5.20(b) and 5.20(c) run against it**.

The two chapters have thus become mirror images of each other, and **within each chapter two clauses give contradictory answers for the same capacity band**.

The FoR model uses "**less than 5 MW**" in all four parallel places, and is internally consistent throughout. The logic is plain: if no meaningful capacity is released for another entity, there is no purpose in withdrawing the full-day right from the original entity.

Accordingly: **there is a strong likelihood that these three departures in the Karnataka draft are unintended**

drafting errors. The divergence in the text can be demonstrated beyond argument; the intention cannot. The Commission should clarify the position before the final notification.

The position of projects at the margin turns on which version survives.

2.4 Route C — change of source

Regulation 9.3

What it provides

An applicant that has been issued in-principle or final grant of Connectivity may, **for the same quantum of Connectivity**, change to another renewable energy source (with or without ESS), in part or in full. The Nodal Agency decides within thirty days (9.3.2).

Three hard conditions

1. Once only, ever. (9.3.4)

2. Six months. (9.3.1) Within six months of the in-principle grant, or six months prior to the effective date of GNA, **whichever is later.**

The corresponding period in the FoR model regulations is eighteen months. The Commission has narrowed the window of defence to a third, and that is an independent choice made in the draft. (Verified)

3. Availability. (9.3.3) For an entity with solar hour access, change of source is permitted **only to the extent that non-solar hour access has not been granted to another entity.**

Which means

This is a route of defence, not a guaranteed right of exit.

Condition 3 makes it **dependent on timing**: if another party has already taken the non-solar capacity at your bay, the route is closed. And condition 1 means it can be used **once** — so there is no sense in holding it in reserve as a last resort; it has to be used in time.

*Read conditions 2 and 3 together: a six-month limit, and capacity that queues. This is the route that closes first.***

2.5 Two routes out — and they are not the same

The draft leaves two separate doors, and which door is yours is determined by your **role**, not by your company. A single company may hold both.

The consumer — Regulation 24.1

One year's notice, together with a fee — **₹50 lakh for distribution licensees and ₹5 lakh for all others** — which is adjusted against the relinquishment charges.

Use of GNA	Relinquishment charges
More than 5 years at InSTS level, or more than 2 years at distribution level	3 months' GNA charges
Less than that	Charges for the shortfall in duration, plus 3 months'

These are not the figures in the FoR model regulations. The FoR model gives 32–33 months; the Commission has brought that burden down to 3 months. That is an independent choice made in the draft, and it is worth recording. (Verified)

Limitation — Interpretive. The draft does not state whether "GNA usage" in Regulation 24.1 is measured **at the date of notice** or **at the date the relinquishment takes effect**. Because the notice period is a year, the difference is large: a distribution-level consumer six months in who gives notice today faces 21 months' charges on the first reading and 3 months' on the second. **The Commission should make the reference point explicit in the final notification.**

The developer — Regulation 23.1

Regulation 24.1 does not apply to generating stations. It applies only to the entities in Regulation 17.1 — distribution licensees, trading licensees, drawee entities and bulk or captive consumers, and standalone ESS drawing

charging power. A solar project is not among them.

The developer relinquishes Connectivity, not GNA — and under Regulation 23.4 the corresponding GNA is reduced automatically.

Thirty days' notice, but a different price:

- **23.2** — Conn-BG1 is encashed; the amount paid towards the terminal bay is forfeited if the bay has been developed or its construction awarded; Conn-BG3 is encashed
- **23.3** — where an ATS is involved, **Conn-BG2 is encashed to the extent of the ATS already awarded or developed**
- On partial relinquishment, BG2 and BG3 are encashed proportionately

And Regulation 16.2 tells you how much that is: Conn-BG2 and BG3 are returned **in five equal parts over five years**. A project exiting two years after commercial operation therefore still has roughly **60 per cent of Conn-BG2 on deposit — and it goes**. The amount of Conn-BG2 is the full estimated cost of the ATS.

What this means

A mature consumer can walk away. The developer is tied in.

For an open access consumer who has completed five years at InSTS level or two years at distribution level, the pressure of January 2027 is essentially **commercial** — the GNA can be relinquished for three months' charges. A consumer who has not yet crossed the threshold is not so free; he pays the charges for the remaining period and three months on top.

The solar developer's pressure is **capital**. His cost of exit equals the amount he has already sunk into the grid, and it **falls with time, not with his decision**.

The consumer's exit cost is determined by how long he has used the network; the developer's, by how much capital is locked in it.

All three routes in this section are therefore for the developer. The consumer has a fourth, and it is cheaper.

2.6 The three routes at a glance

Route	Regulation	For whom	Principal condition	Status
A — ESS within existing Connectivity	5.2 / 5.12	Those with unused headroom in their Connectivity	Nodal Agency approval; Lead Generator responsibility	⦿ Draft + Conditional
B — independent non-solar access	5.10(a) transmission ≥ 5 MW · 5.20(a) distribution < 5 MW	Wind REGS or ESS	Agreement with the bay holder; available capacity (9.2); Conn-BG2 exemption only for a bay built at own cost	⦿ Draft + Verified, implementation Conditional
C — change of source	9.3	Those still within the six-month period	Once only; residual non-solar capacity must remain	⦿ Draft + Conditional

2.7 And one thing common to all three

Eligibility is not entitlement.

All three regulations open a route *to apply*. The actual right to inject depends on sanctioned Connectivity, effective GNA, metering, scheduling, and **available capacity on the network**.

And all three carry the same price — **time**. Route A requires the Nodal Agency's approval, Route B another party's consent, Route C residual capacity. **All three depend on things that queue**.

The window in January 2027 is three months long. **And the draft contains no provision for extending it.**

Section 3 — The slot trap in banking

Position as at September 2026

Two documents, two statuses. The **regulation numbers** in this section are from the **draft**. But the **ToD slots, the tariffs and the DERS structure are from the Tariff Order presently in force** — KERC, 27 March 2025, Annexure-2 and Chapter 6. That order is operative today.

And the banking regime in force does not depend on the draft. Clause 12.e of the Open Access Regulations, 2025 (Gazette, 26 March 2025) already imposes the 8 per cent charge and the same slot asymmetry. **The draft does not create that regime; it fixes a terminal date for it.**

3.1 What Regulation 39 provides

Chapter 8 of the draft, Regulation 39, has five sub-clauses. It carries forward the provisions of the Open Access Regulations, 2025 almost verbatim, with two additions.

39.1 — Duration. Monthly banking is allowed **up to 31 March 2030** from the effective date. The effective date is the date of grant of GNA or the signing of the Wheeling and Banking Agreement ("WBA"), whichever is earlier.

★ **This is the draft's one substantive change, and it is easy to miss.**

Clause 12.e(i) of the Open Access Regulations, 2025 presently confers the entitlement for **five years from the effective date** — that is, separately for each WBA, on a rolling basis. The draft removes that rolling right and substitutes **a single fixed date for all**.

A party executing a WBA in 2027 may today assume a five-year entitlement. Under the draft it stops on 31 March 2030 regardless.

*The residual entitlement therefore shortens with the date of the agreement — a 2027 agreement obtains three years, a 2029 agreement one. **The cost of waiting is built into the draft itself. (Verified)***

And the proviso is the least-read paragraph in this brief: Existing WBAs executed **before 13 January 2023** in the case of short-term open access, and **before 2 January 2023** in the case of long-term open access, **shall not be entitled to the banking facility from the date of renewal**.

For that cohort the cliff is therefore not 31 March 2030. **It falls on their own renewal date.** The quantum is at 3.6.

39.2 — Month-end lapse. Unutilised banked energy remaining at the end of the month **shall not be carried forward** to subsequent months. Renewable Energy Certificates ("RECs") may be obtained for that energy. *(Regulation 43.2 makes separate provision to the same effect for excess injection.)*

39.3 — The slot asymmetry. This is the heart of the trap.

Energy banked in	May be drawn in
Slot-1 and Slot-3	Slots 1, 2, 3 and 4 — all four
Slot-2 and Slot-4	Slot-2 and Slot-4 only

39.4 — Rate. 8 per cent of the banked energy, in kind — "*or as determined by the Commission from time to time.*"

39.5 — Computation. Slot-wise and day-wise, summated at the end of the month, with a worked illustration.

All of the above is Verified — taken directly from the draft text as published in the Gazette.

3.1.1 And one step has been removed from Regulation 39.5

The illustration at Regulation 39.5 is **lifted verbatim from the Commission's own notification** — Notification No. **KERC-7-CON-2022-23/1276/1191 dated 10 January 2024**, "*Methodology of TOD Settlement Procedure in the State of Karnataka*", Tables 1 and 2. The same three days, the same injection and drawal figures (totals of 1,315 and 1,295), and the same banked balance: **135 kWh × 8 per cent = 10.80 kWh**.

But the original notification does not stop there. It goes on: where there is net drawal in peak at month end, it is first to be set off against the net energy banked in off-peak, **and an additional 2 per cent is to be levied on the energy so set off**. In the illustration that is 0.4 units on 20 kWh — **a total of 11.2 kWh**.

The draft stops at 10.80. The final step is absent.

This is not a case of Karnataka adopting a different policy. **The same illustration, the same figures, and one operation removed at the end.** It is established by placing two primary documents side by side. *(The full treatment is at Section 5.) (Verified)*

A minor point, worth recording: the table at 39.5 prints the multiplication as $135 \times 0.08 = 10.08$, while the sentence that follows states 10.80 kWh . The correct figure is 10.80.

3.1.2 ★ But ask one question first — which cohort are you in?

Karnataka does not operate a single banking regime. It operates two, in parallel.

	Legacy cohort	Current cohort
Who	Non-REC projects that applied between 1 April 2018 and the commencement of the GEOA Regulations, 2022	Subsequent WBAs and renewals
Period	Annual	Monthly
Charge	2 per cent of energy injected	8 per cent in kind
Month-end lapse	No	Yes (39.2)
Slot restriction	—	Yes (39.3)
Status	Subject to the outcome before the Hon'ble Supreme Court	In force

Source: Open Access Regulations, 2025, Clause 6 and Clause 12.e. (Verified)

This has a direct commercial consequence: the entire model in this brief is built for the current cohort.

A legacy-cohort consumer bears no 8 per cent deduction, no month-end lapse and no slot restriction. Applying the figures in this brief to such a consumer would be wrong.

The first question before procurement is therefore not "should we buy a battery". It is this:

What is the application date of your WBA, and which cohort does that place you in?

If the answer is not known, establish it first — the economics of the two cohorts are not comparable.

And the draft is silent here. Regulation 39 makes no mention of the legacy cohort. Whether the 8 per cent charge, the month-end lapse and the 31 March 2030 sunset apply to it is not stated. The proviso to 39.1 addresses only renewal.

This is the largest transition gap in the draft, and it is a proper question for the Commission (Section 5.7). (Verified — no such provision appears in the draft)

3.2 The mechanical consequence

The ToD slots, verified from the Tariff Order dated 27 March 2025, Annexure-2:

Slot	Hours	Jul-Nov	Dec-Jun
Slot-1	06:00–09:00	0	+100 paise
Slot-2	09:00–18:00	0	0
Slot-3	18:00–22:00	0	+100 paise
Slot-4	22:00–06:00	0	–100 paise

A separate observation: the ToD differential is nil across all slots from July to November. For five months of the year there is no time-of-use incentive at all.

Now put the two together:

Solar generation falls wholly within Slot-2. Under Regulation 39.3, energy banked in Slot-2 may be drawn only in Slot-2 or Slot-4.

Banked solar energy therefore cannot be drawn in Slot-3 — the evening peak, where the ToD premium is +100 paise.

Two further layers sit beneath this:

First — the asymmetry reproduces the peak / off-peak structure of the Forum of Regulators model exactly. The table in the Commission's orders of 8 June 2023 and 10 January 2024 treats 06:00–10:00 and 18:00–22:00 as

peak, and 10:00–18:00 and 22:00–06:00 as **off-peak**. Mapped onto the present slots, Slot-1 is peak, Slot-2 off-peak (solar), Slot-3 peak, Slot-4 off-peak. **Regulation 39.3 therefore says: what is banked in off-peak may be drawn only in off-peak.** That is verification, not inference.

Second — the restricted band has widened. In the same orders the morning peak ran 06:00–10:00, **four hours**; today Slot-1 runs 06:00–09:00, **three hours**. The daytime off-peak band has thus moved from eight hours to **nine**. *More solar generation now falls within the restricted slot than before — and that tightening was effected by a second, independent instrument.*

3.3 But the conclusion is not "there is no alternative to storage"

This is where most analyses go wrong, and **it is where we were ourselves before running the model.**

The closure of Slot-3 does not make banking worthless. **Slot-4 remains open — and night-time load is real load.**

HT-2(a), FY 2026-27, behind the meter. All figures in rupees per kWh of incremental solar:

Route	₹/kWh
Banking at 8 per cent, where the consumer has Slot-4 load	5.54
Physical storage (discharged into Slot-3)	3.93
Banking at 8 per cent, no Slot-2 / Slot-4 load	0 (<i>Regulation 39.2 — lapse</i>)
Banking at ₹3/kWh (Scenario B)	3.02

Banking is finished for the evening peak, but for night-time load it is worth ₹1.61/kWh more than storage.

Storage prevails in four situations only:

1. The consumer has no load in Slot-2 or Slot-4 — the banked energy lapses and banking is worth nil
2. Demand-charge shaving is credited (₹350/kVA/month; but note the 90 per cent contract-demand floor)
3. Scenario B is accepted — that is, if the banking charge moves from a charge in kind to a charge in rupees
4. After renewal, or after 2030 — when no alternative remains

Three break-evens, and all three are distant:

What would have to change	Present	Break-even
Banking charge	8 per cent	34.7 per cent
Cost of storage	₹2.56/kWh	₹0.67/kWh
Slot-3 ToD adder	₹1.00/kWh	₹4.24/kWh

And this defence matters: the cost of storage is computed on the **lowest rate traced for 2025** — APTRANSCO, December 2025, **₹1.48 lakh per MW per month**. Both IEEFA and JMK have recorded that this sits below commercially viable levels, and have questioned roughly 75 per cent of the two-hour allocation on that basis. At a benchmark of ₹2.3 lakh the gap widens from ₹1.61 to **₹2.81**.

On the most storage-favourable assumption available, storage still loses.

*This does not weaken the brief; it makes it credible. Everyone who says "buy storage" is selling something. The analyst is the one who says "not today — but here is your date."***

3.4 ★ The central finding — they are not competitors

After six iterations of the model, one conclusion did not change across the last three:

Banking and storage are not alternatives to one another. They are two stages in the same queue.

Banking yields more value per kWh, so **fill it first**. Whatever the consumer's Slot-2 / Slot-4 load can absorb goes through banking; **the remainder goes to storage; and what is left after that lapses.**

$$V = \min(u_b, 1) \times ₹5.54 + \min(u_s, 1 - u_b) \times ₹3.93$$

u_b = proportion of incremental solar absorbed in Slot-2 / Slot-4 u_3 = proportion that can be used in Slot-3

The sizing assumption can reverse the conclusion. Show both:

Assumption	Formula	Result
A. As required (default)	$\min(u_3, 1-u_b) \times ₹3.93$	Residual always positive
B. Firm contract — battery contracted in advance	$\min(u_3, 1-u_b) \times ₹6.11 - ₹2.18$	Negative below $u_3 < 35.6$ per cent

The second is the more realistic: a battery is taken on an annual contract, not on monthly requirement. For a low-utilisation consumer, storage therefore destroys value rather than merely adding little.**

The customer map

The value of banking turns on the consumer's **shift pattern**, and on nothing else:

Shift pattern	Slot-3 load	Slot-4 load	Outcome
Three shifts	Yes	Yes	Banking — both slots absorb
Two shifts	Yes	No	Storage — this is the real customer
Single shift	No	No	Neither
Day plus night	No	Yes	Banking

The opening question is one line: *how many shifts do you run?*

Do not ask for utilisation. Ask for the shift pattern. **It is obtained in a single sentence, and it identifies the likely case.**

It does not settle it. Twelve months of 15-minute interval data determines the final answer — and that data is held only by the consumer.

Assumptions — stated

Round-trip efficiency 85 per cent · 2-hour duration · 1 cycle/day · 95 per cent availability · **demand-charge savings taken as nil** · BESS at ₹1.48 lakh/MW/month · HT-2(a) FY 2026-27 energy charge ₹6.60/kWh, demand charge ₹350/kVA · IEX taken as nil (Regulation 5.9(x)(a) closes the contract-free merchant route in Karnataka).

3.5 DERS — a further layer, and it reverses the conclusion

Clause **6.14** of the Tariff Order and **Note 10** to the HT conditions: under DERS, consumption above the base level is charged a **flat ₹5**, and **ToD does not apply to it**. After 31 March 2026 an open access consumer may also opt for DERS.

For such a consumer the evening premium does not exist at all:

	₹/kWh
Banking	4.60
Storage	2.07

The gap widens from ₹1.61 to **₹2.53**.

And there is a commercial consequence worth stating plainly: DERS eligibility turns on consumption above a base level — so the consumers who qualify are, by construction, high-consumption industrial consumers. **On this model, that same qualification weakens the case for storage rather than strengthening it.**

DERS status should therefore be established before any storage economics are run for a consumer. (Verified — Clause 6.14 and Note 10; the inference about who qualifies follows from the clause itself, not from any market observation)

3.6 The renewal cliff — now quantified

The proviso to Regulation 39.1 creates a separate, private cliff ahead of 31 March 2030. **Its size has not previously been measured.**

Source: the Karnataka SLDC Wheeling and Banking list, **dated 9 July 2024** — 556 entries / 4,924.08 MW, reconciled

exactly against the Abstract sheet of that file.

	Agreements	MW
Legacy cohort (executed before 13.01.2023 / 02.01.2023)	358	3,725
GEOA cohort (monthly, 8 per cent)	198	1,199

The legacy cohort is 76 per cent of the W&B fleet.

And within it, the distribution of maturities:

When	Agreements	MW
Peak — FY 2027-28	61	1,024
Between now and 31 March 2030	151	1,948
After 31 March 2030	159	1,275

And after 31 March 2030 not a single agreement in the legacy cohort retains banking — those that escape the proviso are caught by the principal date in 39.1.

There are therefore two cliffs, not one. The first has already begun and peaks in FY 2027-28. The second takes everyone remaining on 31 March 2030.

Limitations — stated

- We do not claim that the old flag in the SLDC file and the cohort under the proviso to 39.1 are **exactly** the same set. Both rest on the same two dates, but the match has not been verified agreement by agreement
- **The denominator is the W&B fleet**, not the State's renewable capacity
- The file is **25 months old**; renewals, terminations and transfers will have occurred since
- For any named project, the WBA, its amendments and the SLDC's current position must be **verified separately**

(Verified, but a snapshot)

3.7 And one group is caught from both sides

Grid constraints in the RTPS belt and the banking cliff are two separate problems — but **they fall on the same map**.

Taking the SLDC's own curtailment notices (7-8 September 2019 · 16 May 2021 · 17 June 2021 · 20 June 2021 · 25 August 2024) together with the W&B list:

In the RTPS belt there are 59 projects / 933.9 MW that lose banking before 31 March 2030.

For context: the belt contains 157 projects / 1,920.6 MW in total, of which the legacy cohort is 123 projects / 1,676.2 MW.

These carry pressure from both directions — the place where energy is curtailed, and the date on which banking ends. **The first genuine demand for storage in Karnataka will come from here.**

Limitation: the notice of 25 August 2024 arose from ICT overload at RTPS and named specific 220 kV substations, which is a locational indicator. But the SLDC notices carry no MW or MU figures — only percentages and durations. This group can therefore be identified but not quantified. (Conditional)

3.8 Conclusion to Section 3

1. The slot asymmetry has extinguished the evening value of banking. **And this is not by reason of the draft — it is in force today**, under **Clause 12.e** of the Open Access Regulations, 2025 (Gazette, 26 March 2025). Draft Regulation 39.3 carries the same structure forward. *The conclusion therefore survives even if the draft does not.*
2. But banking is not dead, and for a consumer with night-time load it remains cheaper than storage. **The recommendation today is not "buy storage".**
3. That changes on one of three dates: **the renewal of your WBA** (for 358 agreements / 3,725 MW), **31 March 2030** (for everyone else), or **if the Commission moves the 8 per cent charge from a charge in kind to a charge in rupees** — which Regulation 39.4 permits.

4. The consumer running two shifts, without Slot-4 load, and not on DERS — **that consumer is a buyer of storage today.** For the rest, the date has not yet arrived.

The advice of this brief is therefore a single line: find your date, and decide eighteen months before it — because all three defensive routes in Section 2 depend on things that queue.

Section 4 — The cost of access

Position as at September 2026

● **Based on the draft.** All regulation numbers are from the Commission's draft (Connectivity and GNA) Regulations, 2026 — Notification KERC/F-32/V-27/2083 dated 4 June 2026, Gazette 11 June 2026. **Where a figure comes from the Tariff Order instead, that is stated.**

The structure of the bank guarantees, their release under Regulation 16.2, and the cost of exit are set out in **Section 2** and are not repeated here.

4.0 Three distinct costs — do not aggregate them

	What it is	When it arises
Entry	Application fees, bank guarantees, cost of the terminal bay	Once, at the outset
Running	The charge stack at 36.3, the monthly charges, the security deposit	Every month, permanently
Patching	T-GNA — temporary access	When the entitlement does not suffice

*The third is the least understood, and it contains the largest trap. Half of this section is devoted to it.***

4.1 Entry cost

Application fees

For what	Level	Fee	Regulation
Connectivity	Transmission — RE, with or without ESS	₹1.0 lakh	3.2
Connectivity	Transmission — non-RE	₹3.0 lakh	3.2
Connectivity	Distribution	₹0.5 lakh	3.2
GNA	Transmission — distribution licensee	₹1.0 lakh	3.3
GNA	Transmission — all others	₹0.5 lakh	3.3
GNA	Distribution	₹0.10 lakh	3.3
T-GNA	Any	₹5,000 per application, non-refundable	27.2

(All amounts exclusive of applicable taxes.)

The T-GNA fee of ₹5,000 is per application — not per MW and not per month. That structure becomes material at 4.4.

The forfeiture ladder — do not confuse it with the bank guarantees

This concerns the application fee, and it is entirely distinct from the encashment of bank guarantees dealt with at 2.5. It escalates in steps:

When	What is lost	Regulation
Minor deficiency not rectified within 5 working days	20 per cent (balance 80 per cent refunded within 15 days)	3.5
Deficiency outside the published list	Application treated as incomplete — rejected, fee returned excluding taxes	3.5
Withdrawal before in-principle grant	50 per cent	3.7
Withdrawal after in-principle grant	100 per cent	3.7
T-GNA application rejected	100 per cent (excluding GST)	27.8

And three further doors close after that:

- **8.2(c) / 8.6(b)** — if Conn-BG1, the cost of the terminal bay and Conn-BG3 are not furnished **within one month** of intimation of in-principle grant, the application is closed and the fee forfeited
- **8.3(e)** — if Conn-BG2 is not furnished within one month, the application is closed, **Conn-BG1 is encashed**, and the fee forfeited
- **10.5** — failure to sign the Connectivity Agreement attracts an extension of not more than thirty days, after which **the final grant itself is revoked**

And the door that opens on the LOA is different. The provisos to 8.2(a) and 8.3(d): once the transmission licensee has issued the **Letter of Award** for the terminal bay or the ATS, withdrawal results in **forfeiture of the amount paid** — not a bank guarantee, but money actually paid over.

*Withdrawal therefore becomes expensive on the date of the LOA, not on the date of commercial operation. That date is not in the developer's hands. (Verified)***

4.2 Running cost

Regulation **36.3** lists the GNA charges — payable both for reserving and for using the network:

	Component	Figure today (Tariff Order, 27 March 2025)
(a)	Transmission charges	Per the order
(b)	Wheeling charges	29 paise/kWh (BESCOM)
(c)	Cross-subsidy surcharge	186 paise/kWh
(d)	Additional surcharge	Not levied — Clause 6.13.6, pending a petition
(e)	Banking charges in kind	8 per cent — Section 3
(f)	Standby charges	125 per cent of the normal energy charge (40.1)
(g)	Losses in kind	Per the order (41.1)
(h)	Other fees and charges	Below

(a) is payable to the transmission licensee; (b) to (f) to the distribution licensee — the Note to Regulation 36.3.

Two small but permanent charges

Regulation **42**: monthly transaction charges of **₹3,000 per month**, and meter reading charges of **₹1,000 per month**.

And Regulation 42(v) contains a concession that is routinely missed: where AMR meters with remote reading are installed and data is transferred automatically to the SLDC or the distribution licensee, **no meter reading charges are collected**. ₹12,000 a year — a small sum, but it goes some way to supporting the case for AMR capex.

Standby — one day's notice is free

The proviso to Regulation 40.1: where standby is required on account of an outage of the generator, and notice is given to the distribution licensee **before the close of the Day Ahead Market on D-1**, the 125 per cent charge does not apply.

This is a provision to build into the O&M schedule. Planned outages are known in advance. (Verified)

Security deposit — and it is not a bank guarantee

Regulation **47.1**: an applicant for GNA shall furnish, as security deposit, an amount equal to **two months' estimated GNA charges — in cash or by demand draft**.

The distinction matters. A bank guarantee is a line; this is cash. **It comes straight off the balance sheet.** And it revolves: 47.2 — proportionate forfeiture on non-payment; 47.3 — **replenishment within 15 days**; 47.4 — cancellation of GNA on failure to maintain it.

The consequences of default are severe: under 46.1, GNA or Connectivity may be discontinued **after fifteen days' notice**, and the SLDC may refuse to schedule and direct disconnection. The third proviso to 36.3 goes further — once GNA is cancelled, **a fresh application must be made**, after clearing the dues.

4.3 The multiplier trap in T-GNA

Now the cost of patching. **And here the draft sets a trap that is not visible on a first reading.**

Regulation 31.3: after the first **three** T-GNA transactions in a financial year, a multiplier attaches to the transmission and wheeling charges —

Transaction	Multiplier
4th	1.25×
5th	1.5×
6th	2.0×
Every one thereafter	2×

And the operative words in the clause are these: "irrespective of period of T-GNA sought."

The multiplier attaches to the **number of transactions**, not to their **duration**.

Which means: a party doing daily or weekly arbitrage — taking T-GNA whenever the market price moves — reaches double charges **within the first week of the financial year**. From the fourth transaction onward he pays a penalty on every one.

Merchant-style short-duration BESS arbitrage is therefore discouraged in Karnataka by the regulation itself. The LOA/PPA requirement at Regulation 5.9(x)(a) was the first door; 31.3 is the second. (Verified)

4.4 The eleven-month structure — the most valuable finding in these Regulations

Regulation 27.1 is a single sentence, and it neutralises 31.3:

"T-GNA may be applied for any period from 1 (one) time block and up to eleven (11) months."

One time block to eleven months. And the multiplier at 31.3 attaches to number, not duration.

Therefore:

Method	Transactions	Multiplier
Taking T-GNA daily	Over 200	2×, from the fourth
A single eleven-month T-GNA	1	None

Take access for eleven months in one application. **One transaction, a ₹5,000 fee, no multiplier — and power may be scheduled every day within that period.**

*The remedy is present in the FoR model regulations themselves, so it will appear in other States. But it is not written down in any published analysis. (Verified)***

Limitation — Interpretive. Regulation 2.1 of the draft **contains no definition of "transaction"** — it defines only "Bilateral Transaction", which is a different thing. Whether "three transactions" means **three grants of T-GNA** or **three scheduled trades** cannot be settled from the definitions.

The text inclines to the first reading: the opening words of 31.3 are **"instead of resorting to avail T-GNA repeatedly"** — the penalty is directed at repeated *availing*; and the parenthesis **"irrespective of period"** appears to have been inserted precisely so that a long T-GNA cannot be counted as several. **But that is a reading, not a definition.**

The decisive document is the one required by Regulation **49.2** — the detailed procedure for T-GNA, to be issued by the SLDC within three months of notification. **Whether the eleven-month structure works will be settled there.**

And a second question sits beside it: the penalty in 31.3 is directed in terms at a **"GNA Grantee"**. Whether an entity with solar hour access seeking T-GNA for injection under Regulation 25.1(c) falls within that description is not stated in the draft.

But read two conditions

1. Getting out costs a month. Regulation 31.2 — an advance T-GNA for more than one month may be reduced for the balance period on **one month's prior notice**, and **the T-GNA charges for that notice month remain payable**.

2. A short T-GNA cannot be revised at all. Regulation 31.1 — an exigency T-GNA, and an advance T-GNA for up to one month, **cannot be revised**. And under any T-GNA, no revision of schedule is permitted **within eight time blocks**.

*The eleven-month structure is therefore cheap but not flexible. It is a decision to be taken at the start of the year, not in the middle of it.***

4.5 But do not treat T-GNA as firm

Giving T-GNA the same standing as GNA in an IRR model would be the most common error available. The draft says in four separate places that it is subordinate:

Regulation	What it provides
28.1	T-GNA is granted only within the Available Transfer Capability — and that, after accounting for the GNA of GNA grantees
28.2	GNA grantees have priority over T-GNA grantees
33.2 / 33.3	The day-ahead schedule for GNA is finalised first; T-GNA takes the balance corridor, and where that is insufficient, pro rata
35.1(a)	In curtailment, transactions under T-GNA are curtailed first, followed by transactions under GNA

And the structure for refusal is strict — **29.2**: where the full quantum and full period sought cannot be granted, **the application is rejected**, unless the applicant has consented in the application itself to a part grant.

Accordingly: apply an availability factor to any revenue attributed to T-GNA. What that factor should be cannot be stated today — **there is no public historical data on ATC. (Unknown)**

*This will change once SOAR is operational — under Regulation 26.1(v) that dashboard is to display corridor availability, pending applications, and T-GNA granted and rejected. Today no one holds that figure.***

One small concession, and it is free: the second proviso to Regulation 21.3(a) — where GNA is yet to become effective, the entity may schedule power up to the quantum of Connectivity, **treated as deemed T-GNA, and shall not be required to pay T-GNA charges**. For the interval between commercial operation and the effective date of GNA, that is a direct cash saving.

4.6 Regulation 25.1(c) — where to sell once the storage date arrives

Section 3 ended with an instruction: find your date. **This is the provision that operates after it.**

Regulation 25.1(c):

- An entity with solar hour access may seek T-GNA **for injection during non-solar hours** — *subject to availability*
- An entity with non-solar hour access may seek T-GNA during solar hours — on the same terms
- Condition: **the total schedule must remain within the quantum of Connectivity**

(The corresponding provision for distribution appears in limb B of Regulation 25.1, by reference to Regulation 5.20.)

This is the only lawful route by which a solar-plus-storage project may sell into the evening after conversion.

Section 1 showed that after January 2027 a solar entity cannot inject during non-solar hours. **Regulation 25.1(c) opens that door again — but not as an entitlement.**

And this is where 4.3 to 4.5 come together:

- The access is a **concession** — *"subject to availability"*
- It is **T-GNA**, and therefore behind GNA under 28.2, and **the first to be curtailed** under 35.1(a)
- Sought repeatedly, it attracts the multiplier at 31.3 — **which makes the eleven-month structure at 27.1**

necessary rather than merely useful

Evening sales from a solar-plus-storage project are therefore a matter of annual planning, not of daily trading.

And they rest on a corridor that is available only to the extent GNA grantees have left it. This line cannot be entered as contracted revenue in a lender's model. **(Conditional)**

4.7 At a glance

Cost	Amount	Regulation	Note
Connectivity application (transmission, RE)	₹1.0 lakh	3.2	
GNA application (transmission, others)	₹0.5 lakh	3.3	
T-GNA application	₹5,000	27.2	Per application — this is where 4.4 comes from
Application-fee forfeiture	20 / 50 / 100 per cent	3.5, 3.7	Distinct from encashment of bank guarantees
Withdrawal after LOA	Full amount paid	8.2(a), 8.3(d)	The date is not in the developer's hands
Wheeling	29 paise/kWh	Tariff Order	BESCOM
Cross-subsidy surcharge	186 paise/kWh	Tariff Order	
Additional surcharge	Nil	6.13.6	Pending a petition
Banking (in kind)	8 per cent	39.4	Section 3
Standby	125 per cent	40.1	Nil if notice is given a day ahead
Monthly transaction	₹3,000/month	42(iv)	
Meter reading	₹1,000/month	42(v)	Nil with AMR
Security deposit	2 months' GNA charges	47.1	Cash / DD — not a bank guarantee
T-GNA multiplier	1.25× → 2×	31.3	From the fourth transaction

4.8 Conclusion to Section 4

1. The small fees do not matter. The structure does. ₹1 lakh or ₹5,000 are rounding on any project. **But the multiplier at 31.3 and the cash at 47.1 show up on the balance sheet.**

2. Two things are outside the developer's control — the date of the LOA (8.2(a), 8.3(d)) and the availability of ATC (28.1). **Both enter the model as assumptions, and there is no public historical data for either.**

3. One sentence may save a substantial sum annually. Regulation 27.1 — a single eleven-month T-GNA, one transaction, no multiplier. **It is written in the Regulations and nobody is using it** — on the reading that one grant is one transaction. *But with "transaction" undefined, the final answer lies in the procedure under Regulation 49.2 — and that procedure is worth making representations on.***

4. And Regulation 25.1(c) answers the question left by Section 3 — but answers it incompletely. Once the storage date arrives, power can be sold into the evening through T-GNA. **But it is a concession and not an entitlement; it is the first to be curtailed; and it must be arranged at the start of the year.**

In one line: the cost of entry for storage in Karnataka is not large. Its uncertainty is — and the data needed to measure that uncertainty will not exist until SOAR is running.

Section 5 — Inconsistencies in the draft

Position as at September 2026

The claim made in this section is deliberately narrow.

A divergence in the text can be demonstrated. **An intention cannot.** In each case below the words of both documents are set side by side; the conclusion is the reader's.

5.0 Two different things — do not run them together

	What it is	Status
An error	The draft contradicts itself — two clauses give different answers on identical facts	Demonstrable
A departure	The draft diverges from the Forum of Regulators model , but remains internally consistent	Demonstrable

The first the Commission will have to correct. **The second is within its powers** — a State Commission is not bound to adopt the FoR model as it stands. But **the direction** of a departure is worth recording.

Both appear below, and they are kept apart.

5.1 The 5 MW mirror — the draft at odds with itself

The draft uses the 5 MW threshold in **six places**: in two chapters, three clauses each.

	Transmission	Distribution
Eligibility	4.5 — ≥ 5 MW → full day	4.8 — < 5 MW → full day
Conversion	5.10(b) — < 5 MW → full day	5.20(b) — > 5 MW → full day
Fresh application	5.10(c) — ≥ 5 MW → solar hour	5.20(c) — < 5 MW → solar hour

(In every instance the threshold attaches to the same thing: whether enough capacity is released for non-solar hours to be made available to another entity.)

The same error in both chapters, running in opposite directions

Transmission: 5.10(b) and 5.10(c) are wholly consistent with one another — *less than 5 MW released → remain full day; 5 MW or more released → become solar hour*. But **4.5 says precisely the opposite**.

A transmission project releasing 5 MW or more is therefore given **full day access** by 4.5 and **solar hour access** by 5.10(c). **The same project, the same facts, two clauses, two answers.**

Distribution: exactly reversed. 5.20(b) and 5.20(c) are consistent with one another — but **both run counter to the FoR direction**, and **4.8 runs counter to them**.

What the FoR model says

The FoR model regulations use **"less than 5 MW"** in all four parallel places, and are internally consistent throughout.

The logic is plain: if no meaningful capacity is released for another entity, there is no purpose whatever in withdrawing the full-day right from the original entity — because the withdrawal benefits no one.

What can and cannot be drawn from this

What can: the three transmission clauses and the three distribution clauses are **mirror images of one another**. Three clauses depart from the FoR text — **4.5, 5.20(b) and 5.20(c)** — and in each case the effect of the departure is the same: **an internal contradiction within that chapter**.

What cannot: whether this is deliberate policy or a drafting error. **No such claim is made in this brief.**

One observation is nevertheless worth recording: a deliberate change of policy does not contradict its own chapter. When policy is changed, all three clauses are changed together. (Verified — as to the divergence in the text; intention Unknown)

Who is affected

Those whose bay releases non-solar capacity in the region of 5 MW.

Such a transmission project, reading 4.5, will assume a full-day right; the Nodal Agency, reading 5.10(c), will grant it solar hour access. **That dispute will surface not at the application stage but at the grant stage** — that is, after the bank guarantees have been furnished.

5.2 ★ The 2 per cent departure — the sharpest finding in this brief

This is not an inconsistency. It is a subtraction.

Where the draft's illustration came from

Regulation **39.5** of the draft contains a worked illustration of the banking computation — a thirty-day table, four slots, charges computed daily.

That illustration is lifted verbatim from the Commission's own notification:

Notification No. **KERC-7-CON-2022-23/1276/1191 dated 10 January 2024** "*Methodology of TOD Settlement Procedure in the State of Karnataka*" — Tables 1 and 2

The same three days. The same injection and drawal figures. The same totals — **injection 1,315 / drawal 1,295**. And the same final banked quantum: **135 kWh**.

But the original notification does not stop there

	Original notification (10.01.2024)	Draft (39.5)
Banking charge at 8 per cent	$135 \times 0.08 = 10.80 \text{ kWh}$	$135 \times 0.08 = 10.80 \text{ kWh}$
Net peak drawal set off against net off-peak banked energy	Present	Absent
Additional 2 per cent on the energy so set off	$20 \times 0.02 = 0.40 \text{ kWh}$	Absent
Total	11.20 kWh	10.80 kWh

The draft stops one step short. That step has been removed.

Why this cannot be described as "a different policy"

Because **the illustration is the same one**.

Had the policy changed, the figures would have changed with it. **Here it is the same three-day table, the same 1,315 and 1,295, the same balance of 135 — and one final operation missing.**

This is established by placing two primary documents side by side. No secondary source is required.

The signed originals are in hand: **the KERC order dated 8 June 2023** (the ToD principle) and **the notification dated 10 January 2024** (the method of computation). (Verified)

The commercial consequence

Note precisely on whom the 2 per cent step falls: **on the consumer in net drawal during peak** — that is, one who consumes in Slot-1 and Slot-3 more than he has banked there.

The step the draft has removed is therefore not one that favoured banking — it was an additional burden upon it.

Under the draft that burden goes. *But banking also receives a terminal date of 31 March 2030 (Regulation 39.1).* **One hand gives and the other takes — and the hand that gives may have opened by accident.**

If the Commission has done this deliberately, the final notification should say so. If it is inadvertent, correction now is in everyone's interest — because the whole economics of banking rests on it.

5.3 Three departures that appear deliberate

These are **not inconsistencies**. The draft is internally consistent here; it simply differs from the FoR model. **And it differs in a way worth noticing.**

#	Subject	FoR model	KERC draft	Direction
1	Window for change of source (9.3.1)	18 months	6 months	Window narrowed to a third
2	Burden of relinquishing GNA (24.1)	32-33 months' charges	3 months'	Burden reduced roughly tenfold
3	Secondary use of GNA (22.1)	To other GNA grantees	Only within the "same group company"	Market closed

The direction is not uniform, and that is the point:

- **#1 and #3 tighten the entry side** — less time to defend, and no facility to lend the entitlement to another
- **#2 loosens the exit side** — a consumer can leave for three months' charges

*The draft therefore makes it harder to come in and to stay in, but easier to leave.***

#3 has a separate consequence: had 22.1 been left open, a market would have formed in unused GNA let out to other entities — and the SLDC portal at 22.3 is being built for exactly that. **The draft retains the portal and removes the market.** In Karnataka that option is closed today. **(Verified)**

5.4 And the chain of internal references is broken in many places

This is not important as a point in its own right. **It bears on the reading of 5.1.**

Internal cross-references and numbering in the draft are wrong in a number of places:

Where	What
5.20(a)	Sub-clauses numbered (iii) and (iv) — there is no (i) or (ii)
8.6	Sub-clauses in the order a) → d) → b)
Proviso to 8.6(a)	Sub-clauses (i), (i), (ii) — the first numeral appears twice
23.5(c)	"clause (a) of Regulation 23.2" — there is no such clause in 23.2; it is 21.3(a)
23.5(c)	"sub-clause (a)(i) of Regulation 23.5" — 23.5(a) has no sub-clauses
23.5(d)	"sub clauses (a) to (d)" — the list runs (a) to (e)
28.6(iv)	"Declarations as provided in Regulation 27.6" — the declarations are in 27.7, not 27.6
49.1.1 / 49.1.2	Both cite "Regulation 5.8/5.18" — the contents of the application are at 5.9/5.19
49.2(a) / 49.2(b)	Both cite "Regulation 27.5" — advance and exigency applications are at 27.4
8.6(a)(i)	"supervision charges under Regulation 12.9" — 12.9 does not concern supervision charges
12.5 / 12.6	One sentence broken across two clause numbers — "In case of an entity covered under" / "12.6 Regulation 4.1 (A), the dedicated line..."
Table at 39.5	Prints $135 \times 0.08 = 10.08$; the following sentence reads 10.80. The correct figure is 10.80

And therefore

In twelve places a number, a cross-reference or an arithmetical result is wrong.

These could be set aside as typographical — **but they bear directly on the reading advanced at 5.1.** Where the internal referencing of a document has failed in this many places, **the likelihood that the reversed 5 MW direction in the same document represents a policy choice is correspondingly reduced.**

That is a line of inference, not of proof — but it is better stated openly than left implied. (Interpretive)

5.5 And one question nobody has been able to put

Regulation	What it provides
1.2	"These regulations shall come into force from 01.10.2026."
54.1	"...the KERC (Terms and Conditions for Open Access) Regulations, 2025 shall stand repealed, from the date of notification of these Regulations."

If the date of notification and the date of commencement differ, which regulations govern the interval?

The old ones repealed, the new ones not yet in force. Regulations 54.2 and 54.3 say nothing about that gap — they provide only that what has been done remains valid, and that subsisting agreements continue.

*This is most probably a drafting slip. But if it is not, the real question is under which regulations applications for banking, open access and T-GNA are to be processed during that period. (Verified — as to the divergence in the text)***

5.6 And one thing that is not an inconsistency

Regulation 27.1 permits T-GNA for any period from one time block up to **eleven months**. Regulation 31.3 imposes a multiplier beyond three T-GNA transactions in a financial year.

These appear to conflict. They do not.

Both operate in the same direction: **make short, repeated transactions expensive; keep long, planned access cheap.** The commercial signal is clear — **planned access, not daily arbitrage.**

Placing this in a list of inconsistencies would be an error. Its proper place is Section 4, where it is presented as an opportunity — with its limitation.

5.7 Six questions the Commission should answer in the final notification

The objection window has closed. The draft was published in the Gazette on 11 June 2026; on the terms of its own preamble, the thirty-day period for objections closed on or about 11 July 2026.

This is therefore not a representation. These are six questions to which the draft still contains no answer, and **the final notification, or the reasoning accompanying it, should address them** — because bank guarantees are being furnished against them now.

(Anyone still wishing to file a formal representation should take advice on whether a late submission remains open.)

The six questions:

- 1. The 5 MW threshold — state the direction.** 4.5 or 5.10(c)? 4.8 or 5.20(c)? *And why are transmission and distribution mirror images of each other?*
- 2. The illustration at 39.5 — has the 2 per cent step been removed deliberately?** If so, will the orders of 8 June 2023 and 10 January 2024 be amended correspondingly?
- 3. "GNA usage" in Regulation 24.1 — measured at the date of notice, or at the date relinquishment takes effect?** *(Section 2.5)*
- 4. "Transaction" in Regulation 31.3 — provide a definition.** Does one eleven-month T-GNA count as one transaction? *(Section 4.4; the answer will fall to the procedure under Regulation 49.2)*
- 5. Regulations 1.2 and 54.1 — which regulations govern the interval between notification and commencement?**
- 6. And what of the legacy cohort?** A Non-REC cohort with **annual banking at 2 per cent** exists under Clause 6 of the Open Access Regulations, 2025, subject to the outcome before the Hon'ble Supreme Court. **The draft makes no mention of it.** Do the 8 per cent charge, the month-end lapse and the 31 March 2030 sunset apply to it — and if so, under what saving or transition provision? *(Section 3.1.2)*

And one broader request: the internal cross-references in the draft (the table at 5.4) should be checked before the final text is settled. **A chain broken in twelve places will generate disputes at the implementation stage** — and by then those disputes will involve projects that have already furnished bank guarantees.

5.8 Conclusion to Section 5

- Among the six clauses using the 5 MW threshold there are **three departures**, and in each chapter their effect is an **internal contradiction**. The text can be demonstrated; the intention cannot.
- The illustration at Regulation 39.5 comes from the Commission's own notification — **the same figures, with the final 2 per cent step missing**. This is the most directly provable point in the brief.
- Three departures appear deliberate — the window at 9.3 narrowed, the burden at 24.1 lightened, the market at 22.1 closed. **Within the Commission's powers, but a direction worth recording.**
- And the chain of internal references is broken in twelve places. **That is not a separate problem — it conditions how point 1 should be read.**

This brief has one message for the Commission: the six questions should be answered in the final notification, because money is being committed against them now.

Section 6 — What other States are doing

Position as at September 2026

This section is not about Karnataka, but it shows Karnataka where it stands.

Sections 3 and 5 showed what the slot asymmetry does. **The question here is different: how have others handled the same difficulty?**

The answer is not simple. **Karnataka is not at the extreme — but in one specific respect it stands alone.**

6.1 First, dispose of a misconception

The slot asymmetry is not a Karnataka invention.

The same structure appears in the Forum of Regulators' model regulations: energy banked in peak may be drawn in either band; energy banked in off-peak may be drawn in off-peak on payment of the banking charge alone. **The balance at month end lapses, with RECs to that extent.**

The structure described in Section 3 therefore derives from the national model. Karnataka did not create it.

But the model contains one further limb

The FoR model permits energy banked in off-peak to be drawn in the peak slot — without the banking charge, on payment of such additional charge as the Commission concerned may determine.

It is precisely that limb Karnataka has removed.

And it did not go at one stroke. **It was removed in stages, and every stage is on record:**

Date	What happened
★ 9 January 2018	KERC order — banking reduced from one year to six months , and in the same order drawal during peak hours limited to energy banked during peak — a possible antecedent of the slot restriction (Conditional) . Challenged before APTEL (<i>Appeal 42/2018 and batch</i>)
8 June 2023	KERC order — ToD settlement introduced. An additional 2 per cent for drawing off-peak energy in peak
10 January 2024	Method of computation notified — off-peak to peak adjustment permitted on payment of 2 per cent
December 2024 draft	KERC proposal — " <i>having regard to grid security</i> ", off-peak energy not to be used during peak hours
26 March 2025	Open Access Regulations, 2025, Clause 12.e(ii) — the 2 per cent route is gone
2026 draft	Regulation 39.3 — the same text carried forward, plus the sunset of 31 March 2030

Six stages, eight years, and a primary document available for each. (Verified)

The claim at Section 5.2 therefore completes itself here: Karnataka removed one particular limb of the national model — and the illustration at Regulation 39.5 carries the imprint of that same subtraction.

6.2 Four States, four different treatments

State / instrument	Slot rule	Charge	Distinguishing feature
Karnataka — OA Regs 2025 + draft 2026	Off-peak-banked only in off-peak	8 per cent in kind	The off-peak-to-peak route has been removed
Gujarat — GERC draft, 6th amendment 2026	The same — off-peak-banked not drawable in peak	Interim ₹1/kWh , but the methodology is prescribed	A cap of 30 per cent of monthly consumption on banked energy
Andhra Pradesh — APERC GEOA 2024, 1st amendment	Three slots. Peak-banked in all three; off-peak-banked in off-peak only; normal-banked in normal and off-peak	—	A grid-level ceiling — 5 per cent of State peak demand
Maharashtra — RE & ESS	Slot crossing permitted in the >100 kW open	Per kWh, on a stated	Banking priced, not prohibited

Policy, GR dated 18 March 2026	access category	principle	
--------------------------------	-----------------	-----------	--

Three distinct approaches emerge, and they are alternatives to one another:

- **Karnataka — prohibition.** The slot cannot be crossed, and price is not considered
- **Gujarat and Andhra Pradesh — a ceiling.** Crossing is limited, but the limit is placed on **quantity**
- **Maharashtra — price.** Crossing is permitted, and it is charged for on a computed basis

6.3 Maharashtra — and it has moved in the opposite direction

The Maharashtra policy sets out a banking table at **Clause 7.1.1**, differentiated by consumer category:

Consumer category	Period	Slot crossing?	Charge
Up to 3 kW	Annual	Yes	None
3–10 kW	Monthly	Yes	Grid Support Charge, per kW
10–100 kW	Monthly	No	Grid Support Charge, per kW
>100 kW (Green Open Access)	Monthly	Yes — across slots	Per kWh banked

That is to say: in the very category with which this brief is concerned — the open access consumer above 100 kW — Maharashtra permits the slot to be crossed, and charges for it.

In Karnataka, energy banked in Slot-2 can never reach Slot-3, at any price.

And the principle for setting the charge — note this

The Maharashtra policy records what the banking charge is to be based upon: **(a) the cost of energy storage, or (b) the difference between the ToD tariffs in the non-solar peak and the solar slot.**

Both of those figures are produced by the model in Section 3 — subject to one important correction of basis:

	Our figure
(a) Cost of storage — per kWh banked	₹2.18/kWh
(b) The ToD differential	₹2.00 (Dec-Jun), ₹0 (Jul-Nov)
Karnataka's 8 per cent charge in kind	Far below either
Our break-even	34.7 per cent

A note on units — this error is easily made. The levelised storage cost in the model is **₹2.56/kWh, but per discharged kWh** — that is, per unit leaving the battery. The Maharashtra banking charge, by contrast, is levied on the kWh **banked** — the unit going in.

At 85 per cent round-trip efficiency, one input kWh yields 0.85 discharged kWh, so the correct figure for comparison is **₹2.56 × 0.85 = ₹2.18 per kWh banked.**

*Both figures are correct; they sit on different denominators. Use ₹2.18 when comparing against the Maharashtra principle. (Verified)***

The Maharashtra policy has adopted precisely the principle that Prayas urged upon the Commission and that the Commission declined.

*That is a policy choice, not an error. But it marks a difference between two neighbouring States, and the difference runs against Karnataka. (Verified)***

And Maharashtra has a storage mandate — but not the one usually reported

This is misreported almost everywhere, and it is worth correcting here:

Provision	To whom	How much
Clause 5.4 — rooftop	Solar/wind above 100 kW	50 per cent of RE capacity, 2 hours
Clause 7.2 — open access	Solar/wind/hybrid above 100 kW	50 per cent of RE capacity, 2 hours
Clause 5.2.1 — DISCOM procurement	Co-located RE procured by the licensee	50 per cent, 2 hours to FY 2029-30; minimum 4 hours from FY 2030-31

△ The "4 hours after 2030" condition appears only in Clause 5.2.1 — for projects procured by the DISCOM.

The open access and rooftop mandates (5.4, 7.2) make no mention of four hours. They provide only for "review every two years".

The trade press has run the two provisions together. A developer planning a 2031 open access project will size on the wrong basis as a result. **(Verified — original Government Resolution dated 18 March 2026)**

And three further points — these are regulations, not policy:

- MERC has already amended the RPO-REC regulations — **RPO of 43.33 per cent and an Energy Storage Obligation of 4 per cent by FY 2029-30. Binding.**
- **Clause 5.1** — transmission charges, wheeling charges, electricity duty and cross-subsidy surcharge do not apply to energy drawn for storage. **Condition: the stored energy must be consumed within Maharashtra.**
- **Clause 7.2.2** — captive Green Open Access, supply within Maharashtra, **minimum 4 hours** of storage → **electricity duty exemption for ten years.** (*Four hours does appear here — and only for captive.*)

6.4 Gujarat — an answer to how the charge should be set

Gujarat's slot rule is much the same as Karnataka's — **energy banked in off-peak may not be drawn in peak.**

But the method of setting the charge is entirely different. The GERC draft derives the banking charge from:

- Actual **15-minute block data**
- Market prices
- Variable cost of generation
- Transmission charges and losses
- **The ratio of net revenue impact on the DISCOM to the energy banked**

The interim figure is **₹1/kWh**, but **the methodology is written down** — which means the figure can be recomputed each year.

And a second provision: **a cap of 30 per cent of monthly consumption on banked energy.**

Karnataka's 8 per cent is a proportion. Gujarat's ₹1 is a price, and there is a method behind it.

The difference is not in the number but in accountability: Gujarat must justify its figure annually; Karnataka need justify the 8 per cent not at all.

6.5 Andhra Pradesh — a third slot, and a ceiling on the grid

The APERC structure has **one slot more** than Karnataka's — peak, normal and off-peak:

- **Peak-banked** → drawable in all three
- **Normal-banked** → normal and off-peak
- **Off-peak-banked** → off-peak only

The middle slot is thus partly flexible, and that is what Karnataka lacks.

But Andhra Pradesh's real distinction lies elsewhere:

A grid-level ceiling on banking — 5 per cent of the State's peak demand.

The SLDC conducts an annual study, and the figure is **notified on 1 March each year.** Capacity commissioned before 2024 is excluded.

This is a direct alternative to Karnataka's "grid security" rationale.

Karnataka gave the security of the grid as its reason and **closed the route.** Andhra Pradesh gave the same reason, **kept the route open and placed a measurable ceiling on it** — and that ceiling is determined by an annual study.

*Both are concerned for the grid. One measures the quantum; the other does not.***

6.6 And in Maharashtra the same fight is running now

(The first version of this sub-section was wrong. It has been rewritten in full after the original order was checked — see 6.6.3.)

6.6.1 MERC has imposed the same restriction as Karnataka

MERC, Case No. 75 of 2025, order dated 25 March 2026.

MSEDCL sought that energy banked during solar hours be drawn back **only in the same ToD slot. The Commission accepted that.**

And the reasoning is worth noting: the Commission held that the banking provision in its own Multi-Year Tariff order of 28 March 2025 was "*not consistent with the DOA Regulations*" — that is, **an "error apparent on the face of the record"** requiring correction.

What the earlier position had been: paragraph 7.13.93 of the MYT order — energy banked in solar hours could be drawn at **any time other than peak hours**. That is roughly **seventeen hours** a day.

The Maharashtra Commission has therefore moved in the same direction as Karnataka — and did so mid-term, by way of review. (Verified — original order, mahadiscom.in)

6.6.2 But APTEL has stayed that restriction

Date	What happened
28 March 2025	MYT order — banking of about 17 hours (para 7.13.93)
25 June 2025	First review order — restriction imposed
3 November 2025	Bombay High Court — order quashed for breach of natural justice; remanded
17 Nov 2025 / 9 Feb 2026	Supreme Court — remanded with a twelve-week deadline
25 March 2026	Fresh MERC order — restriction re-imposed after hearings
24 April 2026	Bombay High Court — writ dismissed, petitioners relegated to APTEL (<i>no view on merits</i>)
★ 27 April 2026	APTEL, interim order: the revised banking dispensation stayed, the invoices issued under it stayed, and MSEDCL restrained from coercive action

The restriction in Maharashtra is therefore not in operation today. The appeals are pending. **(Verified — court judgments and three independent law-firm notices)**

6.6.3 And the "no abrupt change" argument — it is not here, but elsewhere

An earlier draft of this brief stated that MERC had accepted that argument on the authority of a Karnataka case. That was wrong, and wrong in two ways.

First — the passage appears at **paragraph 4.3.8** of the order, and it is **a stakeholder's objection**, not the Commission's conclusion. The Commission's analysis of banking is at paragraphs 18.1–18.22, and the argument is not accepted there. *The High Court in fact recorded that the Commission's analysis contains "absolutely no discussion" of why the solar companies' objections did not find favour.*

Second — the decision cited in that objection **is not a KERC order**. It is a judgment of **APTEL** — *Fortune Five Hydel Projects Pvt. Ltd. v. KERC, Appeal No. 42 of 2018 and 18 connected appeals, dated 29 March 2019*, before Justice N. K. Patil and Shri S. D. Dubey.

(The original judgment has been checked on aptel.gov.in — 167 pages.)

6.6.3.1 And that judgment produces something that changes Section 6.1

The judgment concerns **the KERC order dated 9 January 2018** — Petition Nos. 90/2016, 100/2016, 104/2016, 47/2017 and 130/2017. The ESCOMs had sought reduction of banking from one year to **three months**; the Commission reduced it to **six months**.

But the same order contained a second provision, and it may be the ancestor of today's Regulation 39.3:

"Limit the drawal of energy from the grid during peak Time of Day (ToD) hours to energy banked during the peak ToD hours." — as the order of 9 January 2018 is described in Green Infra's written submissions (para 6.3(b))

The notion of a slot restriction was therefore not born in 2023. It is visible in 2018 — in the very order that cut banking, and it was challenged in the same appeal.

△ **A limitation of wording.** The passage above is **the appellant's description of the order** — not the verbatim text of the KERC order itself. It should therefore not be asserted as "the parent of 39.3".

The safe formulation: *the KERC order of 9 January 2018 contained a restriction linking peak-ToD drawal to peak-ToD banking; it is a possible regulatory antecedent of the present slot structure. The exact continuity remains to be verified against the original order.*

*The chronology at 6.1 is accordingly not five stages but six, running from 2018. (Conditional)***

6.6.3.2 And the same judgment contains a substantial defence for the Commission

The appellants' central argument rested on **PTC India Ltd. v. CERC, (2010) 4 SCC 603**. That Constitution Bench judgment holds:

A **regulation** under Section 178 is **subordinate legislation**, and it **can override subsisting contracts, including power purchase agreements**. An **order** cannot.

In 2018 the Commission acted by order. In 2026 it is acting by regulation.

And that is the route PTC India itself sanctions.

*The 2026 draft is therefore not vulnerable on the ground that made the 2018 order vulnerable. **The brief should say so plainly** — because if it does not, the reader will draw the wrong conclusion, and a lender's counsel will catch it on the first page.*

The surviving argument is narrow but live. In the same appeal (para 5.18) the appellants contended that such an amendment "**has to be prospective in nature**", and should apply only to agreements executed "*post the Impugned Order*".

Karnataka's question in 2026 is exactly that: the sunset of 31 March 2030 in Regulation 39.1 applies to **WBAs already in existence. Whether that is prospective or retrospective is the only real question that remains.**

● **Where verification stops — and it will not go further.** The first **75 pages** of the judgment have been read (*the same 75 on two separate attempts; a limitation of PDF text extraction, not of size — the remainder must be read in a browser*). Those pages contain the case, the impugned order, the eleven questions of law (*No. X — legitimate expectation; No. XI — retrospective effect*) and the parties' arguments. **The Tribunal's own reasoning is in the following ~90 pages.**

6.6.3.3 But the outcome of the appeal is visible in the Commission's own regulations

What the appeal decided can be established without the Tribunal's pages — because the Commission has recorded it in its own regulations.

Open Access Regulations, 2025, Clause 6: Non-REC projects that applied between 1 April 2018 and the GEOA Regulations, 2022 are entitled to **annual banking at 2 per cent** of energy injected — and that entitlement is expressly "**subject to the outcome before the Hon'ble Supreme Court**".

Two things follow, and both from primary documents:

1. That cohort still has **annual banking** — the six-month reduction of 2018 has therefore not held against them. *That the appellants succeeded is admitted by the Commission's own regulations.*
2. But it is **not final** — the Commission has itself made the entitlement subject to the outcome before the Supreme Court.

A step back is required here: the word "subject to" does not establish what stage the appeal has reached. It may be a pending appeal, a consequential proceeding, or simply regulatory caution. Without the court record that stage cannot be stated — and it should not be stated in the brief.

Karnataka therefore operates two banking regimes in parallel today — the legacy regime of **annual banking at 2 per cent** (*subject to the outcome of the proceedings*) and the current **monthly regime at 8 per cent**. The 2026 draft does not mention the first at all.

And the legacy cohort loses that concession on renewal — the proviso to Regulation 39.1. (Verified — OA Regs 2025, Clause 6)

6.6.3.4 The Tribunal's holding — verified from the primary judgment

The full 167-page judgment has now been read. (An earlier version of this brief could not extract pages 75-167 and drew a conclusion from the pages it had. That conclusion was wrong; the correction is at 6.6.3.6.)

The Tribunal framed three issues of its own (page 129):

Issue 1	Whether the impugned order modifying the terms of banking arrangements and concluded contracts, retrospectively , is sustainable in law
Issue 2	Whether it was passed in violation of natural justice, promissory estoppel, legitimate expectation
Issue 3	Whether it was passed without substantial data / analysis and is a non-speaking order

And it held, in terms:

Para 15.14 (p. 157) — "...we are of the considered opinion that the impugned order has been passed by the Respondent Commission in gross violation of the principles of natural justice, doctrine of promissory estoppels and legitimate expectation, etc."

Para 17.1-17.2 (p. 165) — in the absence of supervening public interest, the Commission reduced the banking period "in a pre-matured manner giving the promises / assurances go by before the expiry of the validity period of 10 years guaranteed by the previous orders." The appellants "cannot be left in lurch in the mid-stream", and the tests in **Pawan Alloys** are not satisfied.

Para 17.3 (p. 166) — the order "is not sustainable in the eyes of law... passed without adhering to the principles of natural justice, doctrine of promissory estoppels & legitimate expectation... Besides, the order is not supported by sufficient / requisite data and analysis. Hence, the impugned order is liable for setting aside and the batch of appeals deserves to be allowed."

ORDER (p. 166) — the appeals are **allowed**, and the KERC order dated **09.01.2018** in Petition Nos. 90/2016, 100/2016, 104/2016, 47/2017 and 130/2017 is **set aside**.

All three grounds are the Tribunal's own. (Verified — primary judgment, pp. 129, 157, 165-166)

And the Tribunal's reasoning at 15.13 is worth noting separately: the Commission may alter the conditions of an executed contract at the instance of either party after hearing both sides; here the amendment was made suo motu. The WBAs themselves provided (Clause 13.6 / 12.6) that they could be amended by KERC only after such a hearing.

6.6.3.5 But state the holding narrowly

This matters more than the verification itself.

Do **not** write: "Fortune Five prevents KERC from changing the banking rules."

Write this instead:

In Appeal No. 42 of 2018 and batch (29 March 2019), APTEL set aside KERC's order of 9 January 2018 insofar as it modified the terms of banking arrangements in concluded Wheeling and Banking Agreements. The Tribunal held the order to have been passed in violation of natural justice, promissory estoppel and legitimate expectation, and to be unsupported by sufficient data and analysis. Its reasoning turned on a suo motu amendment of executed contracts without hearing the affected parties — the agreements themselves permitted amendment only after such a hearing — and on the absence of supervening public interest before the ten-year validity guaranteed by earlier orders had expired.

The result turned on the instrument and the process, not on any general immunity from future regulation.

And the distinction at 6.6.3.2 holds from the other direction. In 2018 the Commission acted by **order**, amending concluded agreements suo motu. In 2026 it is proceeding by **regulation**, with a published draft and a thirty-day objection window — the route *PTC India* sanctions, and with the process defect cured.

So the surviving question is narrow, and it is the right one: the sunset of 31 March 2030 applies to agreements **already concluded**. Is that prospective regulation, or the modification of concluded contracts by another name? **On the Tribunal's own reasoning at 17.1-17.2, much may turn on whether supervening public interest is established — which the draft, as it stands, does not attempt.**

6.6.3.6 A correction, recorded

An earlier version of this brief stated that natural justice and promissory estoppel appeared only in the appellants' written submissions (paras 5.29, 5.34) and should not be attributed to the Tribunal. That was wrong.

Those doctrines do appear in the submissions — but the Tribunal **also framed them as Issue No. 2 of its own** (p. 129) and **held on them** at paras 15.14 and 17.3. The error arose because pages 75–167 could not be extracted at the time, and a conclusion was drawn from the pages available rather than from the whole.

*The rule that follows is recorded in the Annexure: where a claim rests on a judgment, the judgment is read to its operative order, or the claim is not made.***

⚠ **What still must not be printed.** The Supreme Court proceeding. **The number Civil Appeal No. 9619 of 2019 has been checked and is wrong** — the court record for that number shows *Anjali v. Lokendra Rathod*, a motor accident matter decided 6 December 2022. **Do not print that number, and do not substitute a guess.** A proceeding appears to exist (*from Clause 6 of the Commission's own regulations*), but its identity and status must come from the court record.

6.6.4 And one further point emerges from this

In Karnataka the route of hearing is open. In Maharashtra it had been closed, and the Court held that to be the breach.

MERC's first review order (25 June 2025) **was quashed for breach of natural justice** — for altering the MYT order without hearing stakeholders. KERC, by contrast, **has published a draft and allowed a thirty-day objection period**, and in January 2026 MERC was required to hold hearings at six centres.

*On procedure, therefore, the Commission's position is strong. A challenge will not rest on process but on substance — on legitimate expectation.***

6.7 Where Karnataka sits

	Prohibition	Ceiling	Price	Must a reason be given?
Karnataka	✓	✗	✗	No
Gujarat	Partly	✓ (30 per cent)	✓ (with methodology)	Yes, annually
Andhra Pradesh	Partly	✓ (5 per cent of peak)	✗	Yes, annual study
Maharashtra — policy (GR 18.03.2026)	✗	✗	✓ (with a stated principle)	Yes, principle written down
Maharashtra — MERC order 25.03.2026	✓	✗	✗	Stayed by APTEL

⚠ **The position in Maharashtra is not settled today.** The State policy permits the >100 kW open access category to cross slots and charges for it; **the MERC order closed that for MSEDCL's consumers**; and **APTEL stayed that order on 27 April 2026.**

Maharashtra is therefore not Karnataka's alternative today — it is Karnataka's own fight, in court. And that is the material difference: in Maharashtra it is being contested; in Karnataka nobody has contested it at all.

Karnataka is not at the extreme — Gujarat's slot rule is as strict, and MERC has imposed the same restriction.

But Karnataka is the only one where no measurable reason need be given for the restriction, and where the restriction has not been challenged. Gujarat's methodology, Andhra Pradesh's annual study, Maharashtra's pricing principle — each has at least one quantitative test. **And in Maharashtra, where there was no such test, APTEL stayed the order.**

Which is why the 8 per cent is difficult to defend: no document states where it came from.

6.8 Conclusion to Section 6

- 1. The slot asymmetry is national, not Karnataka's.** But **Karnataka has removed the FoR model's paid off-peak-to-peak route** — and that same subtraction is visible in the illustration at Regulation 39.5 (*Section 5.2*).
- 2. Three alternative treatments exist, and all three are in operation** — Gujarat's methodology, Andhra

Pradesh's 5 per cent ceiling, Maharashtra's pricing principle. **The argument that there was no alternative does not hold.**

3. Maharashtra's policy moved the other way, but its Commission moved Karnataka's way. The Government Resolution permits slot crossing and prices it; MERC closed that route; **APTEL stayed the order.** *Maharashtra is not the answer — it is a matter still running.*

4. And an appellate judgment on Karnataka itself already exists. *Fortune Five Hydel Projects Pvt. Ltd. v. KERC & Ors.* — APTEL, Appeal No. 42 of 2018 and batch, 29 March 2019 — arose from the Commission's own reduction of banking. Its **outcome is recorded in the Commission's own regulations** (Clause 6, OA Regs 2025), and its **reasoning is corroborated at pages 155-157** by a secondary source citing the primary. **No neighbouring Commission is needed; the authority is on Karnataka directly.**

Its **holding is now verified from the primary judgment** — the Tribunal framed natural justice, promissory estoppel and legitimate expectation as **its own Issue No. 2**, and held on all three at paras 15.14 and 17.3 before setting the order aside.

But state it narrowly. **The Tribunal set aside an order that modified concluded agreements suo motu, without hearing the affected parties; it did not bar prospective regulation.** In 2026 the Commission is proceeding by regulation, with a published draft and an objection window. **The surviving question is therefore whether a sunset applied to agreements already concluded is prospective regulation or the modification of a concluded contract — and, on the Tribunal's own reasoning, whether supervening public interest has been established (6.6.3.5).**

One question can be put to the Commission, and it is not a hostile one: on what basis was the figure of 8 per cent arrived at?

Three neighbouring States have that answer written down.

Annexure — method, sources and limitations

Position as at September 2026

This part is not for readers. It is for those who wish to check.

If any figure in this brief does not persuade you, its origin is here. **And where it is not here, that is stated.**

A. Method and scope

A.1 What this brief examines

Three consequences of the draft Regulations for **open access and captive solar projects in Karnataka**: the conversion of the right to inject during non-solar hours, the slot restriction on banking, and the terminal date for banking. **And when, out of those, the storage decision arrives.**

A.2 What it does not examine

- **It is not legal advice.** The arguments in Sections 5 and 6 should be settled with counsel
- **It is not investment advice.** The model uses an example project; yours is different
- **Project-specific 15-minute load and injection data has not been used** — it is not public, and it makes the largest difference of all
- **The full open access charge stack has not been modelled.** The principal model is for *behind-the-meter / captive self-consumption*. Wheeling, cross-subsidy surcharge and additional surcharge are set out separately in Section 4 and are not blended into the model
- **No procurement negotiation is reflected.** The BESS rate is taken from a public tender, not from a quotation

A.3 Draft and operative law — the distinction is maintained

This distinction is observed throughout, and it matters:

In force today	Still a draft
Open Access Regulations, 2025, Clause 12.e — 8 per cent, slot asymmetry, month-end lapse, the condition on renewal	Connectivity/GNA Regulations, 2026 — Regulation 39, plus the sunset of 31 March 2030
Tariff Order dated 27 March 2025 — ToD slots, tariffs, DERS	—

The core of Section 3 — the slot asymmetry — therefore survives whether or not the draft is notified.

What depends on the draft: the ninety-day window in Section 1, the three routes in Section 2, the T-GNA material in Section 4, and **the date of 31 March 2030.**

B. Status marks — definitions

Mark	Meaning
Verified	Taken directly from the specific clause, table or computation in a primary document
Conditional	The provision exists, but its use depends on contract, technical condition or approval
Interpretive	A conclusion drawn from the text — not stated in the text
Assumption	A model input; if it changes, the conclusion changes
Unknown	The question is posed; no answer is available
⌘ Publication gate	Not to be printed until verified

And **⌘ Draft-based** is not a status mark — it is the status of the document. A clause in the draft may be verified and still not be in force.

B.1 The rule of verification

A claim is "verified" only where its basis can be located in the relevant text, order, table or computation of a primary document.

A name in a bibliography is not evidence.

And in a judgment or an order, a paragraph number is not enough either — the block it sits in has to be identified. Submissions, objections and findings all carry paragraph numbers. *"Ought to be set aside"* is the language of counsel; *"we are of the considered opinion"* is the language of a tribunal.

But that test alone is not sufficient, and this brief learned it the hard way. A doctrine can appear in the submissions and in the findings. **Where a claim rests on a judgment, the judgment is read to its operative order — or the claim is not made.** A partial reading produced a wrong conclusion in v1.2; the correction is at 6.6.3.6.

*Twice in the preparation of this brief a party's argument was taken for a tribunal's holding — once at paragraph 4.3.8 of a Commission's order, and once at paragraphs 5.29 to 5.34 of an appellate judgment. Both would have been printed as reasoning had the surrounding block not been checked.***

*Nor is the first rule abstract. In the course of preparing this brief, it was found repeatedly across three working sessions that a claim had been attributed to a document which could not possibly have contained it — on one occasion a judgment of a tribunal was attributed to a tariff schedule, and one and the same document recurred as the purported source of several unrelated claims. Reading the footnote is not enough; it has to be opened.***

C. Source register

C.1 Primary — regulatory

Document	Identification	Used for
Draft KERC (Connectivity & GNA) Regulations, 2026	Notification KERC/F-32/V-27/2083 dated 4 June 2026 · Karnataka Gazette Part III, Volume 161, Issue 112, dated 11 June 2026	All regulation numbers in Sections 1-5
KERC (Terms & Conditions for Open Access) Regulations, 2025	KERC-2-TR-2024-25/1805 dated 26 March 2025 · Gazette Part III No. 200	Clause 6 and Clause 12.e — the operative banking regime
KERC Tariff Order	dated 27 March 2025 — Annexure-2 (ToD), HT-2(a) tariffs, Clause 6.14 (DERS), Clause 6.13.6 (additional surcharge), Note 10 to the HT conditions	All tariffs in Sections 3 and 4
KERC Review Order	dated 3 March 2026	<i>Checked — it revised tariffs for</i>

		<i>FY 2025-26 only; FY 2026-27 is untouched</i>
KERC order — ToD settlement	dated 8 June 2023	The 2 per cent route — Section 5.2
KERC notification — ToD settlement methodology	KERC-7-CON-2022-23/1276/1191 dated 10 January 2024 , Tables 1 and 2	The source of the illustration at Regulation 39.5
FoR Model Connectivity/GNA Regulations	Forum of Regulators, 99th meeting	Comparison for every departure

C.2 Primary — data

Source	Particulars	Used in
Karnataka SLDC — Wheeling & Banking list	dated 9 July 2024 · 556 entries / 4,924.08 MW · reconciled against the Abstract sheet of that file	Sections 3.6, 3.7
SLDC curtailment notices	7–8 Sept 2019 · 16 May 2021 · 17 June 2021 · 20 June 2021 · 25 Aug 2024 (RTPS ICT overload, named 220 kV substations)	Section 3.7
APTRANSCO BESS tender	December 2025 · 2-hour · ₹1.48 lakh per MW per month	The storage cost assumption

C.3 Primary — other States

State	Document
Maharashtra	RE & Energy Storage Policy 2025-26 to 2035-36 · Government Resolution dated 18 March 2026 , Unique Code 202603181848365810 · Cabinet approval 22 February 2026
Maharashtra	MERC Case No. 75 of 2025 , order dated 25 March 2026 (<i>signed copy on mahadiscom.in</i>)
Gujarat	GERC draft, 6th amendment, 2026
Andhra Pradesh	APERC GEOA Regulations 2024, 1st amendment
Judicial	<i>M/s Fortune Five Hydel Projects Pvt. Ltd. v. Karnataka Electricity Regulatory Commission & Ors., Appeal No. 42 of 2018 and batch, APTEL, judgment dated 29 March 2019</i> (<i>aptel.gov.in</i> ; pp. 1–75 read directly, pp. 155–157 via a secondary source citing them) · Bombay High Court, 3 November 2025 and 24 April 2026 · Supreme Court, 17 November 2025 and 9 February 2026 · APTEL interim order, 27 April 2026

C.4 Secondary — and used only as signposts

The Prayas Energy Group's submission to the Commission (the ₹3/kWh in Scenario B); the Prayas study of banking rulings, which cites *Fortune Five* at pp. 155–157 and quotes the language of the holding; the cautions of IEEFA and JMK on BESS viability; and law-firm notices (three independent ones for the APTEL stay).

One exception to the rule below, stated openly: the reasoning in *Fortune Five* is carried on the Prayas study rather than on the primary judgment, because pages 75–167 could not be extracted. **It is marked Conditional in the text for that reason** (6.6.3.4). Every other conclusion in this brief rests on a primary document.

No conclusion in this brief rests on any of these. They were used to **locate** documents, not to **establish** claims.

D. Model assumptions — all in one place

Parameter	Value	Note
Consumer category	HT-2(a) Industrial	Tariff Order
Energy charge, FY 2026-27	₹6.60/kWh	Tariff Order
Demand charge	₹350/kVA/month	Tariff Order
DERS rate	₹5/kWh, no ToD	Clause 6.14
Banking charge — Scenario A	8 per cent in kind	Regulation 39.4
Banking charge — Scenario B	₹3/kWh	Prayas submission
Round-trip efficiency	85 per cent	Assumption
Duration · cycles · days · availability	2 hours · 1/day · 365 · 95 per cent	Assumptions
BESS capacity charge	₹1.48 lakh/MW/month	The lowest rate traced for 2025
Demand-charge savings	Nil	<i>Deliberately conservative — note the 90 per cent contract-demand floor</i>
Incremental solar, example project	10 lakh kWh/year	Variable

D.1 A caution on units

The levelised cost of storage is ₹2.56/kWh — but per *discharged* kWh.

At 85 per cent round-trip efficiency one input kWh yields 0.85 discharged kWh. **Per kWh banked it is therefore ₹2.18.**

Both are correct; the denominators differ. Use ₹2.18 when comparing against a banking charge — because the banking charge is levied on the unit going in.

D.2 The one assumption that reverses the conclusion

Only one: how much of the incremental solar is absorbed in the consumer's Slot-2 and Slot-4.

At 100 per cent, banking wins. At 0 per cent, banking is worth nothing (Regulation 39.2 — lapse). *Every other assumption is secondary to this, and it is the same point the customer map at 3.4 makes.*

E. Limitations — consolidated

#	What	Where
1	The W&B file is 25 months old ; the denominator is the W&B fleet, not the State's renewable capacity	3.6
2	The file's Old flag and the cohort under the proviso to 39.1 have not been matched agreement by agreement	3.6
3	The curtailment notices carry no MW or MU figures — the group can be identified but not quantified	3.7
4	"GNA usage" in Regulation 24.1 — the draft does not say whether it is measured at notice or at effect	2.5
5	"Transaction" in Regulation 31.3 is undefined; the answer will fall to the procedure under Regulation 49.2	4.4
6	No public historical data on ATC — no availability factor can be placed on T-GNA revenue	4.5
7	The three 5 MW departures — the divergence in the text can be shown; the intention cannot	5.1
8	The 2018 peak-ToD restriction comes from the appellant's description , not from the original order	6.6.3.1
9	Resolved in v1.3. The full 167-page judgment has been read; the Tribunal's holding is verified from the primary at pp. 129, 157 and 165–166. <i>An earlier version wrongly declined to attribute natural justice and promissory estoppel to the Tribunal, having read only pages 1–75. The correction is recorded at 6.6.3.6</i>	6.6.3.4
10	The judicial status of the legacy cohort's entitlement is unverified . "Subject to the outcome" does not establish the stage of the proceeding; and the case number is unknown — the commonly cited CA 9619/2019 was checked and found to be wrong	6.6.3.3
11	The model is for the current cohort only. It does not apply to the legacy annual-2 per cent cohort, which requires a separate base case	3.1.2

E.1 And one false trail, closed here

🔒 **An incorrect citation — not to be used.**

The reference Civil Appeal No. 9619 of 2019 circulates as the Supreme Court proceeding in BESCOM v. Fortune Five Hydrel. **It has been checked. It is wrong.**

The 9619 in the Supreme Court URL is a **diary number**, not a Civil Appeal number; and the judgment attached to it is **Anjali & Ors. v. Lokendra Rathod & Ors.** — a motor accident compensation matter decided on 6 December 2022. **It has nothing to do with BESCOM, Fortune Five, or the banking appeal.**

*A proceeding before the Supreme Court probably exists — the words "subject to the outcome" in Clause 6 of the Open Access Regulations, 2025 suggest as much. But its true number and current status must be taken from the court record.***

This is recorded here for one reason: the reference is in circulation. If the brief simply omits it, the next analyst repeats the error; if the brief records it, the error is made once.

And the MERC matter is kept as context only in this brief — it is live, APTEL has stayed it, and **no conclusion here rests upon it.**

F. What would sharpen this — and who holds it

Three of the five below are not public. If you hold any of them — particularly project-specific interval data — it

would move this brief from estimate to measurement, and I would be glad to say so.

1. **The original KERC order dated 9 January 2018** — to establish continuity with Regulation 39.3
2. **An updated W&B list** from the SLDC, for 2026
3. **Project-specific 15-minute load and injection data** — to move the model from example to reality
4. **A commercial quotation for BESS**, in place of a tender rate
5. **The SOAR dashboard** — Regulation 26.1(v); once operational, the ATC figure becomes available for the first time

G. And a request

This brief is free, and it may be wrong.

If any figure, any clause or any conclusion appears to you to be incorrect, **say where it fails**. Corrections will be recorded, with attribution, in the next version.

And if you hold any of the five items above — particularly project-specific data — it would move this brief from estimate to measurement.

End of Report No. 1.